LIVEWIRE
FOR LIVE CAREERS

Data Science

Day-wise syllabus

Duration: 76 Hours

S. no	Topic
1	Data Science Initials • Introduction to Data Science • Analytics Methodology • Problem Definition • Data Collection • Types & Dictionary Programming with R • Introduction to R programming • Need and Necessity of R in DS • Features of R programming (Variables/Loops/Functions/.etc.) • Case Study (Practicing and Lab work)
2	Programming with R • Features of R programming continue • Data Visualization using R (Inbuild package) • Case Study (Practicing and Lab work) Statistics • Basics of Statistics • Descriptive Statistics 1. Measures of Central Tendencies 2. Measure of Dispersion 3. Measure of Position 4. Measure of frequency • Case Study (Practicing and Lab work)
3	Statistics • Inferential Statistics • Correlation • Covariance • Confidence Interval • Bias • Outlier • Skewness and Kurtosis • Probability Distribution 1. Normal Distribution 2. Binomial Distribution 3. Poisson Distribution • Case Study (Practicing and Lab work)
4	Statistics • Test of Significance 1. Type I and Type II error 2. One tailed two tailed 3. T-Test 4. Chi-Square test 5. Anova test

	• Case Study (Practicing and Lab work) Data Pre-processing • Importance of Data Processing • Client Expectation and Machine Expectation • Steps of Pre-processing (Data quality assessment, Data cleaning, Data transformation, Data reduction) • Case Study (Practicing and Lab work)
5	Machine Learning • What is ML? • Importance of ML. • Types of learning • Supervised learning vs Unsupervised learning • Supervised learning algorithms 1. Classification(Naïve Bayes/SVM/KNN) • Case Study (Practicing and Lab work)
6	Machine Learning • Supervised learning algorithms 2. Regression(Linear Regression/Random Forest) • Unsupervised learning algorithms 1. Clustering(K-means/Hierarchical clustering) • Case Study (Practicing and Lab work)
7	Analytics • Data Analytics overview • Types of Data Analytics • Data Visualization using Ggplot for Decision making • Forecasting analytics • Case Study (Practicing and Lab work)
8	Analytics • Microsoft Excel for Data Analytics • SQL 1. SQL Server Overview 2. MS SQL vs Oracle SQL 3. SQL Server Installation 4. SQL Server Architecture 5. SQL Server management studio (SSMS) • Case Study (Practicing and Lab work)
9	Analytics • SQL 6. SQL Server queries and procedural steps
10	Analytics • SQL "+" Data Science • Data Wrangling using SQL • Social Network Analysis 1. Graph terminology 2. Gephi software installation and use

	• Case Study (Practicing and Lab work)
11	Analytics • Social Network Analysis 3. Creating Network Graph and analyzing them • Data Mining 1. Evolution of Database Technology 2. Data Mining Overview 3. KDD Process in Data Mining • Case Study (Practicing and Lab work)
12	Analytics • Data Mining 1. DM Technique Association Rule Mining 2. Web Mining (Web content, Web Usage and Web Structure Mining) • Case Study (Practicing and Lab work)
13	Data Analytics • DataMining 3. Spatial Mining (MBR and Quad Tree) 4. Temporal Mining (Neural Network with Keras and Tensorflow) • Case Study (Practicing and Lab work)
14	Data Analytics • Data Mining 5. Text Mining (NLP process for Text Mining) • Case Study (Practicing and Lab work) • Project Assignment Status
15	Industry 4.0 • What is Industry 4.0 • Getting Started with AWS (Amazon Web Servies) • AWS security and security group in IAM Role • AWS Features • AWS Architecture • Evolution of Cloud Platform • What is Big Data • Big Data Tools Available in AWS • Why AWS use for Big Data? • Basics of Hadoop • What is Data Warehousing • Introduction to Amazon Redshift
16	Industry 4.0 • Terminology AWS 1. VPC 2. Amazon Lambda 3. GLUE 4. Amazon RDS 5. Data Lake (S3) 6. EC2 7. EMR (Elastic MapReduce) 8. Endpoints

	9. Crawlers 10. Sagemaker 11. Quicksight Integration 12. EFS 13. Amazon Cloud Watch 14. Amazon KMS • Case Study (Practicing and Lab work)
17	Industry 4.0 • Integration of Microsoft SQL with AWS • Creating basic process in AWS using Data Lake and Database SQL • Integration of Big Data with AWS • Case Study (Practicing and Lab work)
18	Industry 4.0 • Creating R Amazon Lambda programs • Working with Data Warehouse • Case Study (Practicing and Lab work)
19	Industry 4.0 • Working with GLUE Studio to stick created process • R programming for Hadoop • Case Study (Practicing and Lab work)